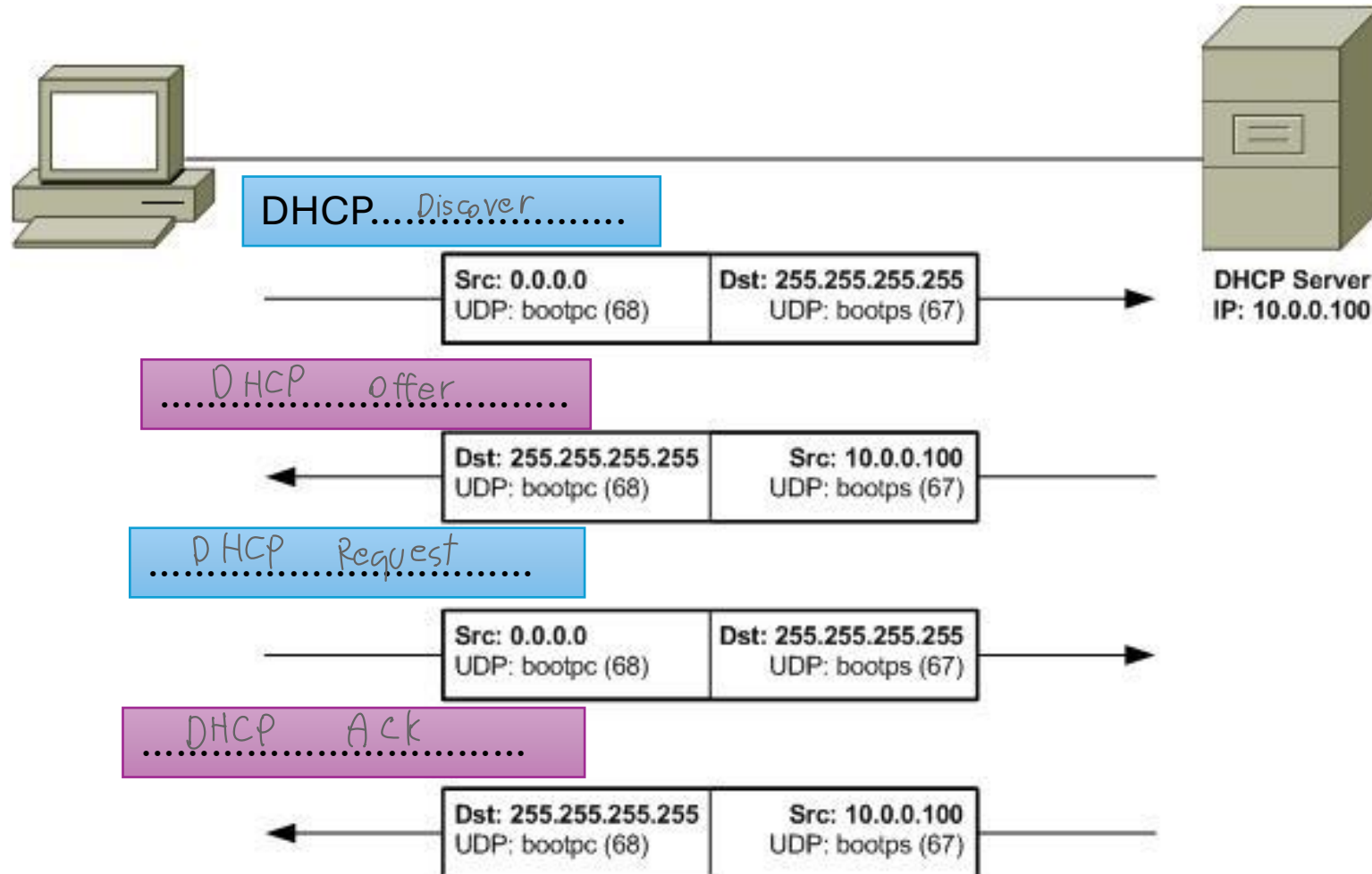


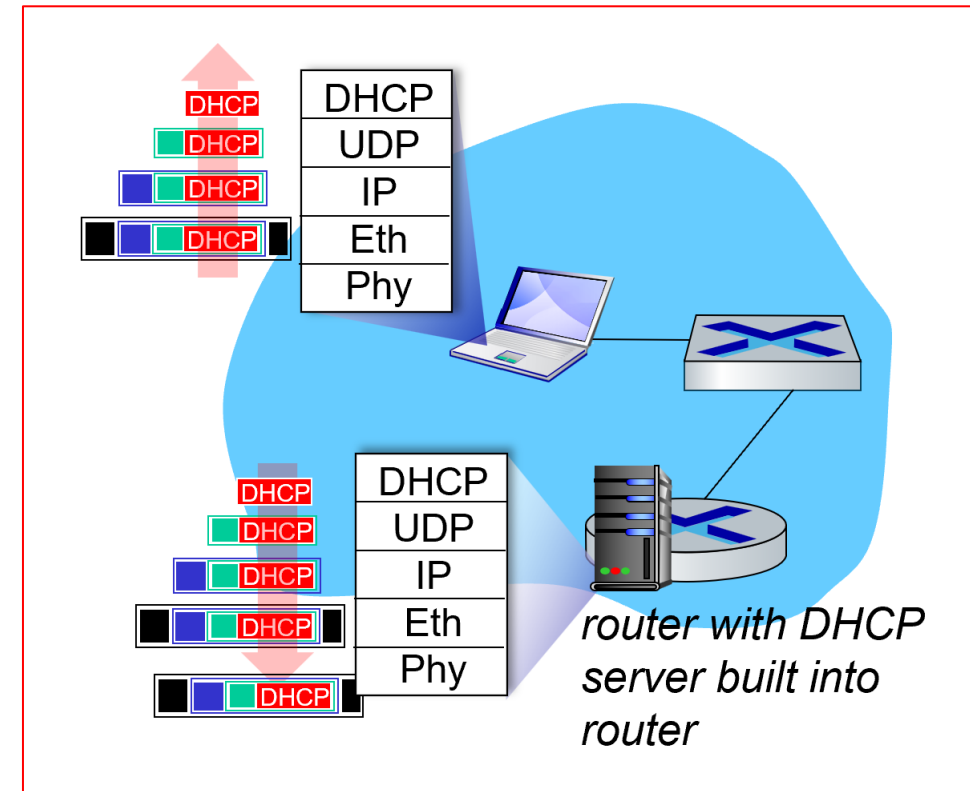
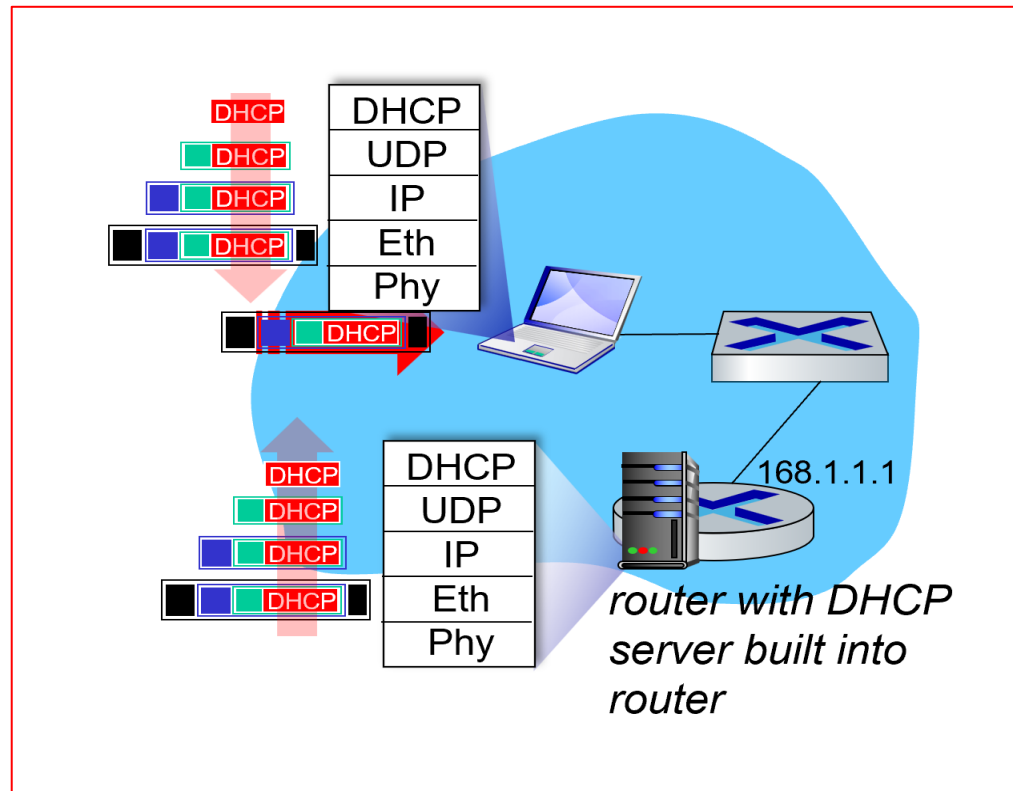
Exercise 1: Recall DHCP Messages

4.1 What are the 4 messages : answer in order.



Exercise 2: Recall DHCP Messages

2.1 From DHCP 4 messages : Which ones are in each direction?



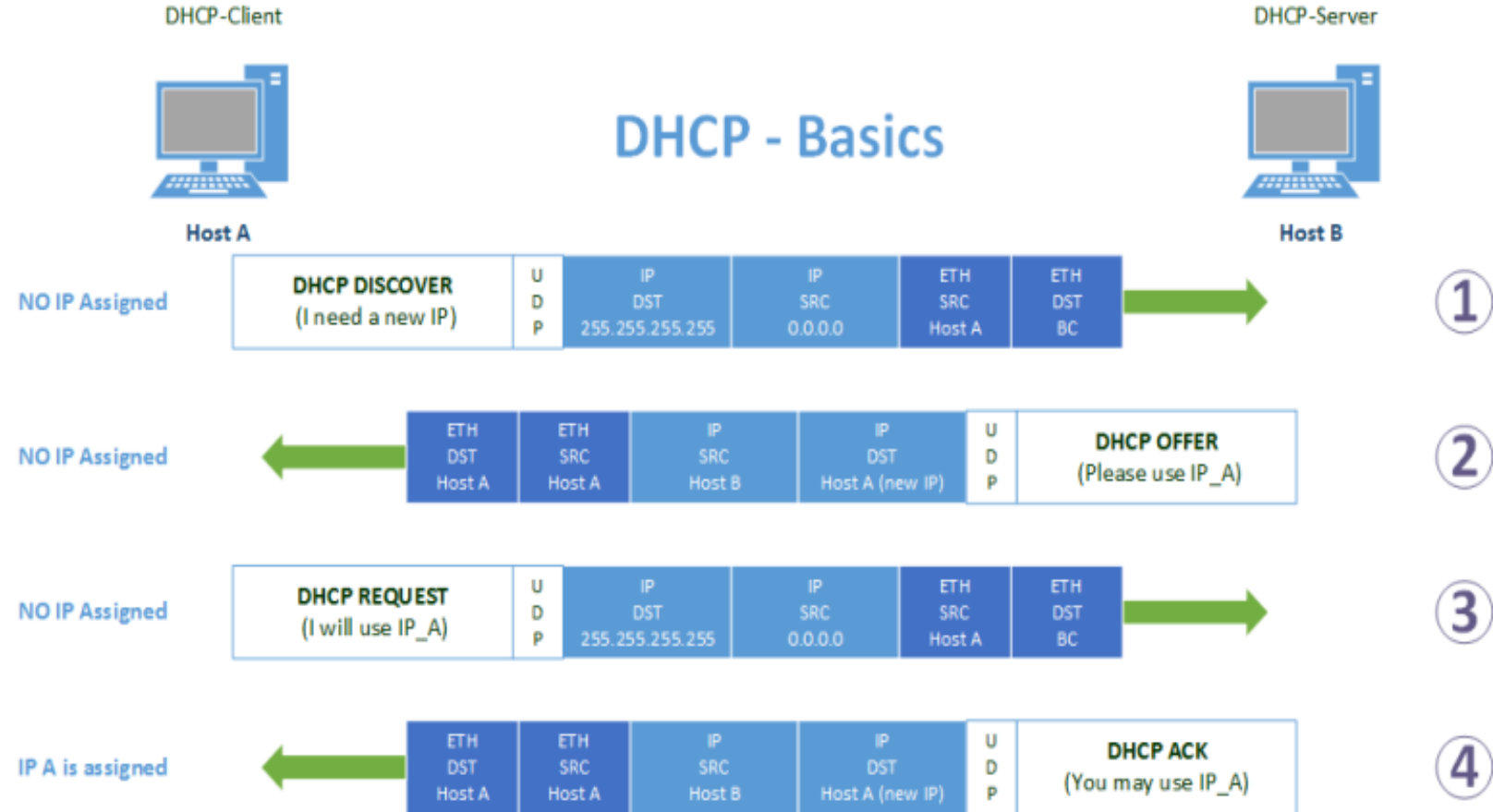
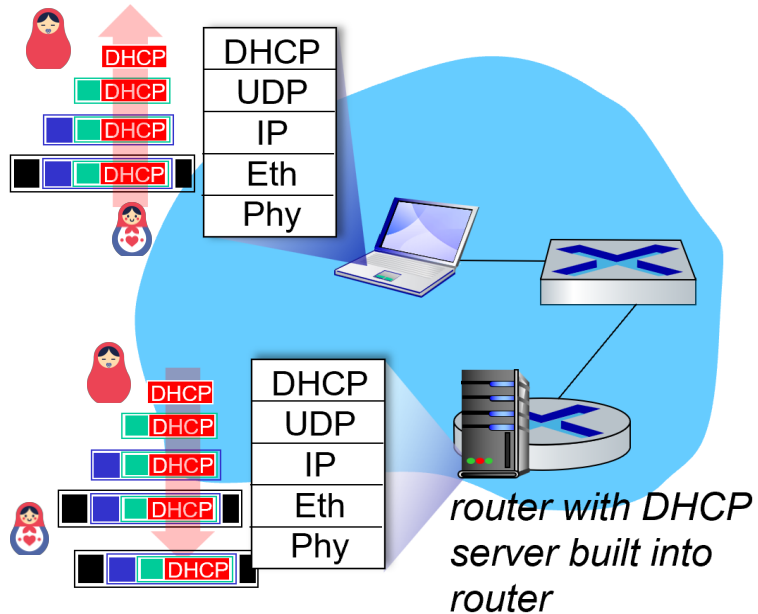
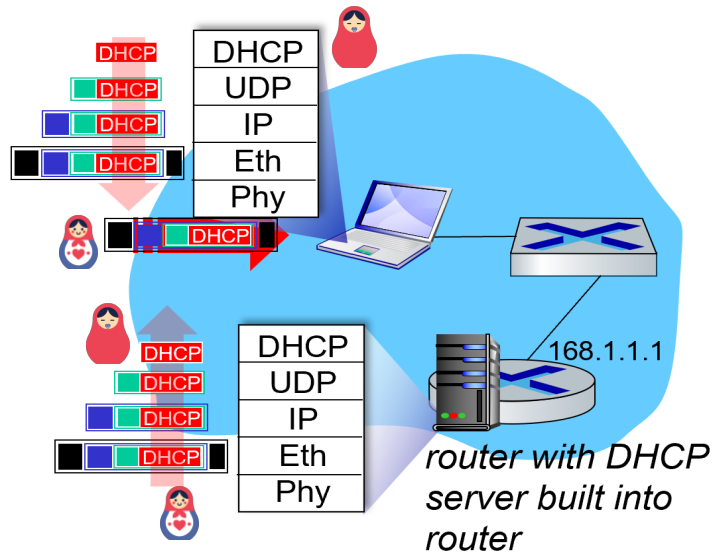
The DHCP messages in this direction can be:

- offer
- Ack

The DHCP messages in this direction can be

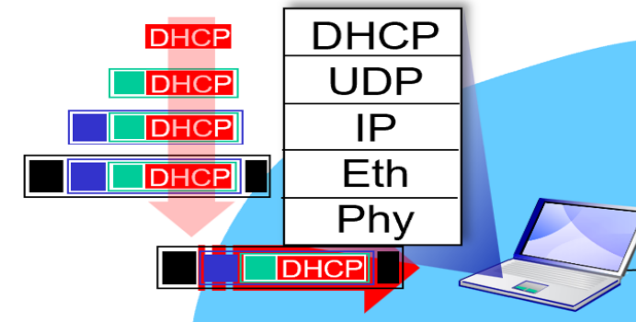
- Discovery
- Request

Exercise 3.1: Headers of each protocol layer in brief



Header of Transport layer:UDP.....
 Header of Network layer :IP.....
 Header of Data link layer:ETH.....

Exercise 3.2: Headers of each protocol layer



```
> Frame 12: 392 bytes on wire (3136 bits), 392 bytes captured (3136 bits) on interface 0
> Ethernet II, Src: Centrale_08:0c:bb:80 (d0:76:58:0c:bb:80), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67
```

```
> Bootstrap Protocol (Discover)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x2a4d077a
  Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: Centrale_08:0c:bb:80 (d0:76:58:0c:bb:80)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
```

```
> Option: (53) DHCP Message Type (Discover)
> Option: (61) Client identifier
> Option: (57) Maximum DHCP Message Size
> Option: (60) Vendor class identifier
> Option: (12) Host Name
> Option: (55) Parameter Request List
> Option: (82) Agent Information Option
> Option: (255) End
```

• รูปนี้แสดง DHCP message รวมทั้ง Header ใน layer ที่ Capture ได้ ให้นักศึกษาระบุว่า ส่วนไหนเป็น

a) DHCP Message (วงเลขคี่)

(53)

b) Header ของ Data link layer พร้อมระบุข้อมูล

ethernet

```
> Ethernet II, Src: Centrale_08:0c:bb:80 (d0:76:58:0c:bb:80), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
```

c) Header ของ Network layer พร้อมระบุข้อมูล

IP

```
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67
```

d) Header ของ Transport layer พร้อมระบุข้อมูล

UDP

```
> User Datagram Protocol, Src Port: 68, Dst Port: 67
```

```
> Bootstrap Protocol (Discover)
```